INCH-POUND

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MILITARY SPECIFICATION

# DOUGH PROOFING BOXES, STEAM HEATED (STEAM, OR STEAM AND WATER, HUMIDIFIED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of dough proofing box.

1.2 Classification. The dough proofing boxes shall be the following sizes as specified (see 6.2):

1.3 Part or identifying number (PIN). The PIN shall consist of the basic number of the specification, and the appropriate PIN structure as specified in 6.7.

| Size | For use with Garrison Bakery Size No. | For use with Central Pastry Kitchen Size No. | For use with Naval Shipboard Installation |
|---|---|---|---|
| 1 | Bak-A-M | BA-6 | ---- |
| 2 | Bak-B-M | BA-3, BA-5 | ---- |
| 3 | Bak-C-M | ---- | ---- |
| 5 | Bak-G-M | BA-7 | ---- |
| 6 | ---- | ---- | ---- |
| 7 (see 6.1) | ---- | ---- | ---- |
| 8 | ---- | ---- | (see 3.6.6) |
| 9 | ---- | ---- | (see 3.6.6) |
| 10 | ---- | ---- | (see 3.6.6) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| A-A-422 | - | Pan Sets, Bread |
| A-A-50110 | - | Rack, Bread, Bakers': 10 Shelf, Knockdown Type |
| QQ-C-576 | - | Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed or Machined Edges, (Plate, Bar, Sheet, and Strip) |

MILITARY

| | | |
|---|---|---|
| MIL-P-514 | - | Plate, Identification, Instruction and Marking, Blank |
| MIL-B-2316 | - | Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-130 | - | Identification Marking of U.S. Military Property |
| MIL-STD-461 | - | Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference |
| MIL-STD-462 | - | Electromagnetic Interference Characteristics, Measurement of |
| MS 20615 | - | Rivet, Solid-Universal Head, Brass, Copper, Nickel-Copper Alloy |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)



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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS

29 CFR Part 1910 - Occupational Safety and Health Standards

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprint of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

BAKING INDUSTRY SANITATION STANDARD COMMITTEE (BISSC)

Standard No. 10 - Prefabricated Enclosures and Air Conditioning Equipment for Fermentation, Proofing, Cooling and Retarding.

(Application for copies should be addressed to the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Chromium-Nickel Steel Plate, Sheet, and Strip, Stainless and Heat-Resisting
A 176 - Steel Plate, Sheet, and Strip, Chromium, Stainless and Heat-Resisting
B 36 - Brass Plate, Sheet, Strip and Rolled Bar
B 117 - Salt Spray (Fog) Testing, Standard Method
B 121 - Leaded Brass Plate, Sheet, Strip and Rolled Bar
B 209 - Aluminum and Aluminum-Alloy Sheet and Plate
B 221 - Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes
B 308 - Aluminum-Alloy 6061-T6 Standard Structural Shapes, Rolled or Extruded

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)



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UNDERWRITERS' LABORATORIES, INC. (UL)

Standard No. 73 - Motor-Operated Appliances

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Standard product. The dough proofing boxes delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with this specification.

3.3 Codes and standards. The dough proofing boxes shall comply with the applicable requirements of UL Standard No. 73, BISSC Standard No. 10, and Occupational Safety and Health Standards No. 29 CFR Part 1910.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the dough proofing boxes he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 73, BISSC Standard No. 10, and Occupational Safety and Health Standards, 29 CFR Part 1910, as follows:

3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be a UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the dough proofing boxes have been tested and conform to UL Standard No. 73.

3.3.1.2 BISSC. Acceptable evidence of meeting the requirements of BISSC shall be:

(1) A photostatic copy of the current BISSC certificate of authorization for the dough proofing boxes, or



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(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the dough proofing boxes are being procured, indicating that the units have been tested and conform to the applicable requirements of BISSC Standard No. 10.

3.3.1.3 Occupational safety and health standards. Acceptable evidence of meeting the occupational safety and health standards shall be a certificate of compliance, certifying that the dough proofing boxes furnished conform to Occupational Safety and Health Standards No. 29 CFR Part 1910.

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dough proofing boxes provided the completed item complies with all provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.4.1 Brass sheet. Brass sheet shall conform to ASTM B 36 or B 121.

3.4.2 Copper.

3.4.2.1 Copper sheet. Copper sheet shall conform to QQ-C-576.

3.4.2.2 Copper rivets. Copper rivets shall conform to MS 20615.

3.4.3 Stainless steel. Stainless steel sheet shall conform to any type steel in the 300 or 400 series of ASTM A 167, or A 176, respectively.

3.4.4 Aluminum. Aluminum shall conform to type 6061 or 6063 of ASTM B 209, B 221, or B 308.

3.5 Design. The dough proofing box shall consist of a paneled frame covered with material that will protect against corrosion and equipped with steam system. Doors shall be provided at two opposite ends, except sizes 5, 6, 7, 8, 9, and 10 (see 3.6.6), which shall have doors at one end only. Rack accommodation shall be as specified in table I.

3.5.1 Air conditioning and humidification. The dough proofing box shall be equipped with an air processing unit for recirculating air in sufficient quantities to maintain a dry-bulb temperature range of 90°F to 110°F and a relative humidity range of 80 to 90 percent within the box. Unless otherwise specified (see 6.2), outside design temperatures shall be 70°F dry-bulb, 53°F wet-bulb. The unit shall be designed to proof a full load of 20 ounce loaves with the quantity of racks shown in table I in 50 minutes.



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TABLE I. Dough proofing box requirements

| Dough proofing box size 1/ | No. of bread racks per box/per aisle | Doors: 2 pass through | Doors: end door | No. of 20 oz loaves per proofing box using bread pans |
|---|---|---|---|---|
| 1 | 4/2 | X | | 800 3/ |
| 2 | 6/2 | X | | 1200 3/ |
| 3 | 10/2 | X | | 2000 3/ |
| 5 | 3/1 | | X | 600 3/ |
| 6 | 2/1 | | X | 400 3/ |
| 7 | 2/1 | | X | 320 3/ |
| 8 | (see 3.6.6) | | X | 200 4/ |
| 9 | (see 3.6.6) | | X | 60 4/ |
| 10 | (see 3.6.6) | | X | 30 4/ |

1/ See 6.3.

2/ Racks conforming to A-A-50110.

3/ Bread pans in accordance with A-A-422, four pan set.

4/ Bread pans in accordance with A-A-422, five pan set.

Each box shall be equipped with an automatic control system that is adjustable to selected points within the specified operating range and shall be capable of maintaining the air processed in the proofing room with a maximum dry-bulb temperature differential of $\pm$ 2°F, and a maximum relative humidity differential of not more than $\pm$ 2-1/2 percent on either side of the point to which the controls are set. Manufacturer's test data shall prove to the satisfaction of the contracting officer that the assembled box and air processing unit meet the requirements of this specification.

3.6 Construction.

3.6.1 Protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wetted in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, plating, isolation, insulation, gaskets, or other means. Fasteners and fittings shall be of the same materials as the materials being joined or supported.

3.6.2 Box.

3.6.2.1 Frame. The frame shall be constructed of structural steel or aluminum shapes and shall be self-supporting before panels are applied. All steel used in the frame shall be hot dip galvanized to conform with the requirements of 3.6.4. When galvanizing is removed by fabrication, it shall be restored to its original galvanized condition. All aluminum used in the frame shall conform to the requirements of 3.4.4.



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3.6.2.2 Panels. All panels shall be fabricated of at least the equivalent thermal conductivity of 1 inch of insulating board, having a maximum K-factor of 0.36 Btu per hour per square foot per 1 inch thickness per $^{O}F$ temperature difference. The inner panel surface shall be constructed from nominal 0.025 inch stainless steel in the 300 or 400 series with a 2B, or better, finish. The outer panel surface shall be steel or aluminum sheet. Steel sheet shall be not less than 0.20 inch thick and shall be galvanized to conform with the requirements of 3.6.4 and then finished in white baked synthetic enamel or stainless steel. Aluminum sheet shall not be less than .032 inch thick. The covering sheets and insulating board shall be bonded together with waterproof cement in accordance with the manufacturer's commercial practice. The insulating materials shall be of non-moisture absorbing materials such as foam rubber, foam plastics, or dried pressed fiber insulation board rendered impervious to moisture. The panels shall be butted together with a mastic sealer, molded rubber, or plastic seal to be air and moisture tight. The interior and exterior panel surfaces for size 8 shall be constructed of stainless steel (see 3.6.6).

3.6.2.3 Doors. Doors shall be not less than 1 inch thick and designed in accordance with the manufacturer's standard product. Provision shall be made to prevent moisture from entering the insulation. The door surfaces shall be constructed from the same materials specified for the panels in 3.6.2.2. Suitably located metal guards or bumper plates shall be provided on both sides and edges of the doors to protect door panels from damage by racks entering or leaving the proofing box. Doors shall be hung with hinges designed and mounted to insure permanent alignment and shall be provided with latches or catches to insure tight closing. Means shall be provided for opening doors from the inside. All hardware, including hinges, exterior bumper bars, and latches shall be finished as specified in 3.8.

3.6.2.4 Door opening. Clear openings for rack entrance shall be not less than 34-1/2 inches wide by 78 inches high for sizes 1 through 6, and not less than 34 inches by 76 inches for size 7. For sizes 8, 9, and 10 the door opening shall be as specified in 3.6.6.

3.6.2.5 Rack guide rails. The dough proofing box shall be provided with one rack guide rail on each interior sidewall and other guide rails spaced between 30-1/2 and 31-1/2 inches apart between rails. The guide rails shall be the only means for guiding the rack through the box. The guide rails shall be 1/4 inch thick plate or equivalent angle or pipe construction supported at each end and at necessary intermediate points above the floor. The height of the guide rails shall be between 8 and 9 inches. Rack guide rails for sizes 8, 9, and 10 dough proofing boxes shall be as specified in 3.6.6.

3.6.2.6 Guard rails. Guard rails, consisting of either 1-1/2 inch galvanized or 2 inch schedule 40 aluminum pipe rail fittings, and 3 inch channel sections shall be provided at the outside corners and side walls to protect the box from contact with racks, pans, trucks, and other rolling equipment.



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3.6.2.7 Electrical requirements. Unless otherwise specified (see 6.2), all items shall be designed for operation on a nominal 208 volt, 60 hertz 3 phase system. Motors or control system of 1/3 horsepower (hp) or less may be supplied for operation on a nominal 120 volt, 60 hertz, single-phase (ac) system. Thermal overload protection shall be required on all motors. Electrical requirements for sizes 8, 9, and 10 shall be as specified in 3.6.6.

3.6.2.8 Lighting. A sufficient number of lights shall be installed within each box to provide a general lighting level of not less than 20 foot-candles when tested as specified in 4.5.4.

3.6.2.9 Inside clear height. The inside clear height of the proofing boxes shall be not less than 78 inches for sizes 1 through 6 and 76 inches for size 7. Each box shall be of sufficient size to accommodate the number of racks specified in table I, each rack being approximately 70 inches long, 29 inches wide, and 72 inches high, except the rack used in size 7 shall be 43-1/4 inches long, 28 inches wide, and 70-1/2 inches high.

3.6.2.10 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with part 10, class C3, group I of MIL-STD-461 (see 4.5.3).

3.6.3 Air processing equipment. The air processing unit shall consist of a conditioning chamber in which heat and moisture are supplied to the air under controlled conditions, and a centrifugal air mover. The air processing chamber and air mover or fan may be incorporated in a common unit casing or in separate units. In addition, an air cleaning system may be incorporated and shall be a direct water type or a recirculating water type with water pump. Air processing equipment shall meet the test specified in 4.5.1.

3.6.3.1 Air processing unit. The casing of each air processing unit and all parts coming in contact with free water shall be fabricated from metal that has been tested for corrosion-resistance as specified in 3.6.4. The casing shall be fabricated of copper sheet specified in 3.4.2.1; brass sheet specified in 3.4.1 not less than 0.027 inch thick; or cold rolled corrosion-resisting steel sheet specified in 3.4.3, not less than 0.030 inch thick. Casing shall be supported and stiffened by structural shapes, constructed of the same type material or material with a corrosion-resistant finish, welded, riveted in place with rivets specified in 3.4.2.2, or bolted in place with brass screws and nuts. Aluminum alloy sheet not less than 0.030 inch nominal thickness also may be used. When aluminum is used, it shall be riveted in place with commercial brazier aluminum rivets and proper means shall be used to prevent electrolytic corrosion between aluminum and copper or brass parts. All water and steam pipe and fittings inside the conditioning chamber shall be brass or copper. External pipe and fittings shall be of the material normally used by the manufacturer for this type of application.



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3.6.3.2 Humidification. Humidification shall be accomplished either by direct steam injection into the airsteam within the unit or by steam injection in conjunction with a direct or recirculated water spray chamber. The design and capacity of the humidification section shall be such as to insure at least 90 percent saturation of the air leaving the unit. Water spray system shall be direct or shall be provided with a direct connected pump, capable of handling the water required for the spray nozzles. A drip pan shall be provided to catch any leaks from the packing glands. Means shall be provided to prevent overflow of the drip pan and resultant damage by either an open or automatic float-valve actuated drain-valve. Brass spray nozzles of the large orifice swirling nonclogging type which produce both an air cleaning and humidifying coarse spray shall be used.

3.6.3.3 Eliminators. Eliminators shall be provided when required for the proper functioning of the equipment furnished. The eliminators shall prevent passage of any particles of spray from the spray chamber.

3.6.3.4 Water sump. When a recirculating water spray system is provided, a watertight sump and an overflow pipe of sufficient capacity to prevent flooding shall be provided. A filter shall be provided over the pump suction intake to prevent passage of solid matter of a size that would clog the pump or damage the impeller. A float-valve with seamless copper ball or plastic float shall be provided for maintaining the water in the sump. The fresh water inlet shall be above the maximum overflow level by at least 1-1/2 times the diameter of the inlet line to prevent siphonage.

3.6.3.5 Heating. Each heating coil shall be of the extended surface type, constructed of copper coated steel or copper tube and shall be provided with fins with cast iron, copper, or brass, sheet or tube headers, or similar approved construction. The trap controlling the air heater may be drained into the spray washer sump when the addition of condensate will not affect the control of air processing. All heating equipment shall be designed for operation with a steam pressure of 10 to 15 pounds per square inch gage (psig). All external pipe and fittings shall be of a material normally used by the manufacturer for this type of application.

3.6.3.6 Fans. The fan shall be of the centrifugal, multiblade type, connected to a standard motor of sufficient size to operate the fan at the required capacity under the actual usage conditions of the system and under conditions of continuous operation. The blower wheel shall be constructed of noncorrosive material or material properly treated to conform to corrosion-resistant requirements specified in 3.6.4. The fan shall recirculate evenly throughout the box a volume of air equal to 90 cubic feet $\pm$ 10 percent per minute per rack. Compliance with the volumetric requirements shall be determined when tested as specified in 4.5.2.



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3.6.3.7 Air distribution. The air distributing system shall provide means for circulating the air evenly within the proofing box (see 4.5.2). All ductwork shall be constructed of galvanized steel or aluminum not less than 0.026 inch thick, reinforced to protect against damage. Distributing ducts shall be protected against corrosion on inner surfaces of ducts. Drop ducts shall be fabricated from not less than 0.0625 inch thick aluminum sheet. The openings for circulating air within the box shall be of such arrangement and size as to produce an even distribution of air throughout the box.

3.6.3.8 Control system. Each control system shall be fully automatic and designed for controlling the temperature of the air discharging into the box. The control systems shall be coordinated to compensate for change in temperature and humidity to produce the desired design temperature and humidity condition. The thermal bulb for control of the dry-bulb and wet-bulb temperature shall be located in a position in contact with the average condition of air within the proofing box and shall be completely protected from any influence other than that of the air processed.

3.6.3.9 Auxiliary manual control. Means shall be provided for manually operating and controlling dry- and wet-bulb temperatures in the event the automatic control system becomes inoperative.

3.6.3.10 Indicators. The equipment shall be provided with temperature, humidity, and steam indicator(s) or any other indicators necessary in the design of the proofing box. The indicator(s) shall be located to permit easy reading and shall provide for calibration and adjustment.

3.6.4 Corrosion-resistance. Material treated to resist corrosion shall not show evidence of corrosion after being subjected to the 48-hour salt spray test specified in 4.4.2.2.

3.6.5 Guards. All rotating, reciprocating, or projecting parts so located as to endanger any person that may come in close proximity thereto shall be enclosed or guarded. All steam lines and appurtenances exceeding a temperature of 150°F. and so located as to endanger personnel or create a fire hazard shall be guarded or covered with insulating material.

3.6.6 Size 8, 9, and 10 for Naval shipboard installation. For Naval shipboard installation the size 8, 9, and 10 dough proofing boxes shall be of sectional design for assembly and erection aboard ship, using simple tools. Each section shall be of a size that will pass through a hatch or door measuring 26 inches wide by 66 inches high or long. The interior and exterior panel surfaces shall be fabricated of type 302 stainless steel specified in 3.4.3, and not less than 0.0375 inch thick.

3.6.6.1 Size 8. The dough proofing box shall be equipped with automatic electric heating and humidity controls. Internal distribution of heat and humidity shall be through integral ductwork. Each proofing box compartment shall be provided with a minimum of ten shelves, five stationary and five adjustable. The stationary shelves shall be spaced six inches apart on



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center. The adjustable shelves shall be on a rack assembly having guide rails that are spaced 1 inch apart on center. The shelves shall be fabricated of stainless steel and measure 24 inches by 72 inches by 1 inch and constructed so that each shelf shall be capable of supporting three 18 by 26 inch sheet pans loaded with rolls, or four 5-strap bread pans filled with 20-oz bread loaves. The dough proofing box shall be designed for connection to a 440 volt, 60 hertz, 3 phase power source. The overall dimensions of the dough proofing box shall be 72 inches high by 80 inches wide by 38 inches deep, and have a clear door opening of 74 inches wide. Overall dimensional tolerance are +0, -1/2 inch.

3.6.6.2 Size 9 and size 10 undercounter dough proofers. A size 9 proof box shall be a double cavity proof box. A size 10 proof box shall be a single cavity proof box. Each cavity shall be equipped with one proofer heater assembly as specified in 3.6.6.2.1. Each cavity shall be capable of supporting a minimum of six 5-strap bread pans as specified in A-A-422. Each cavity shall have a minimum of three removable rack guides and three heavy duty nickel plated racks. The rack guides shall be mounted 6-1/2 inches apart on center. The racks shall be 28-3/8 inches long by 29 inches wide. The interior dimensions of a single cavity shall be 30 inches wide by 30 inches deep by 26 inches high. The overall dimensions of a size 10 single cavity proof box shall be not more than 36 inches wide by 36 inches deep by 34 inches high. The overall dimensions of a size 9 proof box shall be not more than 66 inches long by 36 inches deep by 34 inches high. A temperature indicator shall be mounted in the exterior of the doors enclosing each proofing cavity. The proofer shall be leg mounted on adjustable legs, and the legs shall have an adjustment range from 5 inches high to 8 inches high.

3.6.6.2.1 Proofer heater assembly. For undercounter proofers, removable proofing heater assemblies shall be installed with the capability to produce humidified or dry heat. Each proofer heater shall be all stainless steel construction with a 4 quart water pan reservoir. The top of the flue riser attached to the heater assembly reservoir shall not exceed 24 inches in height. The proofer heater assembly shall be equipped with a 3 position flip toggle switch: ("Heat" - "Off" - "Humidity Proof"). On "Proof" operation, the unit shall automatically maintain a temperature range of $90^{\circ}$ to $100^{\circ}$ and 95% relatively humidity. The proofer shall be connected to a 120 volt, 60 hertz, single phase power source. The unit shall have a thermostat dial control knob and a timer dial control knob The entire proofer heater assembly must be capable of being easily removed from the proofing cabinet with simple tools.

3.7 Lubrication. Provisions shall be made for the proper lubrication of all moving parts. Seals shall be provided where necessary to prevent leakage of lubricant. At time of delivery, the unit shall be completely lubricated with the manufacturer's standard lubricant.

3.7.1 Lubricating chart. The supplier shall attach to each dough proofing box a lubricating chart indicating all points on the equipment requiring lubrication, grade of lubricant, and time interval when lubrication should be accomplished.



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3.8 Finish. Unless otherwise specified herein, proofing boxes shall be finished in accordance with the manufacturer's standard practice, except the color of painted finish shall be white enamel. Hardware, including hinges, exterior bumper bars, and latches shall be galvanized or otherwise suitably protected against corrosion to conform with the requirements in 3.6.4.

3.9 Marking for identification. The dough proofing box shall be marked in accordance with MIL-STD-130 on a photo-sensitive anodized aluminum plate conforming to MIL-P-514. The plate shall be located on the upper right-hand corner opposite the door-opening side. When two door-opening sides are furnished, the plate may be attached at either end.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification.

3.10.1 Fabrication. All sharp burrs, slivers, or splinters shall be removed.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be drawn tight.

3.10.3 Welding. Welded joints shall be complete, uniform, sound, and thoroughly fused, with no pits, undercuts, cracks, burn-holes, or fractures. All scale and flux deposits shall be removed. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall develop strength adequate to the parts connected. Welding process shall be in accordance with a welding code acceptable to the bureau or service concerned.

3.10.4 Electrical assembly. Wiring shall be properly joined, not cut or abraded, not loose at terminal ends, properly enclosed in conduit where required, provided with adequate slack, and not stripped of required insulation.

3.10.5 Solder. All surfaces to be soldered shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

3.10.6 Metal forming. All metal forming shall be done with appropriate tools to insure uniformity of size and shape. Sheet metal or structural sections used in fabrication of the boxes shall be free of kinks and bends. Shearing and chipping shall be done accurately and cleanly and shall be neatly finished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein.



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Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.



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4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made for the following to establish that no deviation is made from requirements specified in 3.6.2.2:

a. Application of paint grip to outer panel.
b. Insulating material application.
c. Panels are butted.
d. Provision for making insulation waterproof.

When a deviation from requirements is noted, correction shall be made to items and process.

4.4.2.2 Salt spray tests. Components shall be subjected to a 48-hour salt spray test in accordance with ASTM B 117 to determine compliance with corrosion-resistance requirements of 3.6.2.1, 3.6.2.2, 3.6.3.1, 3.6.3.6, 3.6.4, and 3.8. The lot shall consist of all like components offered for inspection at one time. The sample unit shall be a single component. Whenever a deviation is noted, correction shall be made to items and process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of dough proofing boxes. The sample unit shall be one dough proofing box. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Not finished in accordance with manufacturer's standard commercial product | 101 | |
| | Galvanized coating removed by fabrication | 102 | |
| | Painted finish not white | 103 | |
| Construction and workmanship, general (applicable to all components and assemblies) | Part missing or not specified type | 104 | |
| | Any component fractured, split, punctured, dented, creased, deteriorated, bowed, sprung, or malformed | 105 | |
| | Part loose | | 201 |
| | Part not in proper alignment | 106 | |
| | Sharp burr, sliver, or edge | 107 | |



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TABLE II. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Classification Minor |
|---|---|---|---|
| Construction and workmanship, general (applicable to all components and assemblies) (cont'd) | Functioning component that is inoperative or does not operate as intended | 108 | |
| | Adjustable assembly that cannot be adjusted | 109 | |
| | Adjustable assembly not properly adjusted to perform the intended function | | 202 |
| | Component not readily accessible for servicing, where required | 110 | |
| | Joint not protected against electrolytic or galvanic corrosion | 111 | |
| Welding and brazing (where required) | Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused | 112 | |
| | Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed | | 203 |
| Soldering | Not adherent or incomplete, not clean (flux residue), not smooth, or pin holes in solder | | 204 |
| Threaded fasteners | Missing, broken, stripped, fractured, or loose | | 205 |
| Electrical assembly | Wiring cut, abraded, not properly joined, loose at terminals, or not enclosed in conduit (where required) | 113 | |
| | Adequate slack not provided to relieve strain or required insulation stripped from wiring | 114 | |
| Gages and instruments | Not specified capacity or type | 115 | |
| | Inoperative or otherwise defective affecting use (visual determination) | 116 | |
| | Not mounted correctly, but will properly function | | 206 |
| Lubrication (where applicable) | Not lubricated where required | 117 | |



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TABLE II. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Assembly | Component not properly assembled or secured | 118 | |
| | Unit perceptibly out of square or alignment | 119 | |
| Identification marking | Missing, incomplete, not legible, not specified type, not in proper location | | 207 |

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of dough proofing boxes. The sample unit shall be one dough proofing box. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. Each box shall be tested in accordance with 4.5. When a first article is not required, the initial production unit shall be tested as specified in 4.5.1, 4.5.2 and 4.5.4, and 4.5.3 when applicable. At the option of the Government any or all tests may be requested if there is a change in design, material, construction, or other departure subsequent to approval of the initial production unit. When the box is to be installed outside the continental United States (see 6.2) each box shall either be tested as specified above, or the supplier shall furnish evidence that a box identical to that offered has successfully passed the tests specified in 4.5.1, 4.5.2, and 4.5.4., and 4.5.3 when applicable, in the manufacturer's plant or in a commercial application. The Government reserves the right to check-test each box to determine the validity of the evidence produced.

4.4.6 Standard compliance. Proof of compliance with the BISSC, UL, and Occupational Safety and Health Standards requirements of 3.3 shall be submitted to the Government representative.

4.4.7 Packaging inspection. The sampling and inspection of the preservation, packaging and container marking shall be in accordance with the requirements of MIL-B-2316.

4.5 Methods of inspection.

4.5.1 Operation. Air processing unit controls shall be connected to a power supply as specified in the contract or order and a water supply to determine conformance with 3.6.3. Steam system shall be leak-tested with



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water at a minimum of 40 psi for 30 minutes. Any pressure drop is cause for rejection of the item.

4.5.2 Air distribution. Air distribution shall be determined for compliance with 3.6.3.7 by measuring air velocities in a completely assembled box with air-processing system mounted on its permanent location but not necessarily connected to utilities except for electric power. Air velocity shall be measured at the discharge of each supply air duct. Measurement shall be made by an appropriate instrument of best commercial accuracy. The test shall be recorded as a failure if any of the following conditions exist:

a. The air velocity of any duct is below or above the limit specified in 3.6.3.6.

b. The air velocity of any duct varies by more than 8 percent of the highest velocity of any duct.

4.5.3 Electromagnetic compatibility test. When electromagnetic compatibility is required, the first article or initial production unit, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report, and the requirements meet MIL-STD-461.

4.5.4 Foot-candle test. The lighting level shall be measured using a Weston Illuminator Meter model 756 or equivalent to determine compliance with 3.6.2.8. Nonconformance with 3.6.2.8 shall constitute failure of test.

5. PACKAGING

5.1 Levels A, B, and Commercial. Preservation, packaging, and packing and marking, for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The dough proofing boxes covered by this specification are used to proof dough in warm, moist air in a uniform manner prior to baking. Size 7 is used with the following items:

MIL-R-43814; Rack, Mobile, Bakery Products and Meat; with Shelves
MIL-O-43815; Oven, Baking and Roasting, Roll-Through, Convection, Electric
MIL-O-43816; Oven, Baking and Roasting, Roll-Through, Convection, Gas



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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size proofing box to be furnished (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.5).
e. Levels of preservation and packing (see 5.1).
f. Outside design temperature if other than specified (see 3.5.1).
g. When electrical characteristics are other than specified (see 3.6.2.7 and 3.6.6).
h. When electromagnetic compatibility is required (see 3.6.2.10).
i. When installation is outside continental U.S. (see 4.4.5).
j. When supplier is to furnish supervision of installation or demonstration service of items. If required, the location, approximate date, the person or office to report to, and any accommodation or service to be supplied by the ordering agency will be stated (see 6.4).

6.2.1 Contract data requirements. Any requirements for equipment manuals for the item covered by this specification should be included in DD form 1423: Contract Data Requirements List, and cited in the contract.

6.3 Sizes. Sizes of bakeries and pastry kitchens specified in 1.2 are in Table of Allowances 50-911, Department of the Army. The bakeries are covered in section III, part 7 for Garrison Bread Bakeries and the pastry kitchens are covered in part 8 for Central Pastry Kitchens. Size 8 is used for Naval shipboard installation.

6.4 Supervision. The following information should be included in procurement documents. When specified (see 6.2), supervision of installation shall be required of the supplier for dough proofing boxes to be erected in the continental U.S. The supplier shall furnish the services of a competent superintendent who shall be familiar with the construction, erection, operation, maintenance, and repair of the equipment and the required utility connections. Material required for complete installation other than specified herein, and all installation labor, will be furnished by others. All utilities will be extended and connected to the equipment by others. The superintendent shall supervise the installation of the equipment and all utility connections thereto, using the material as stated above. Supervision of utility connections shall include all work from points within 10 feet of the dough proofing box. Upon completion of the work, the superintendent shall turn the equipment over to the contracting officer or his representative in proper operating condition and shall instruct personnel designated by the contracting officer in the proper operation.



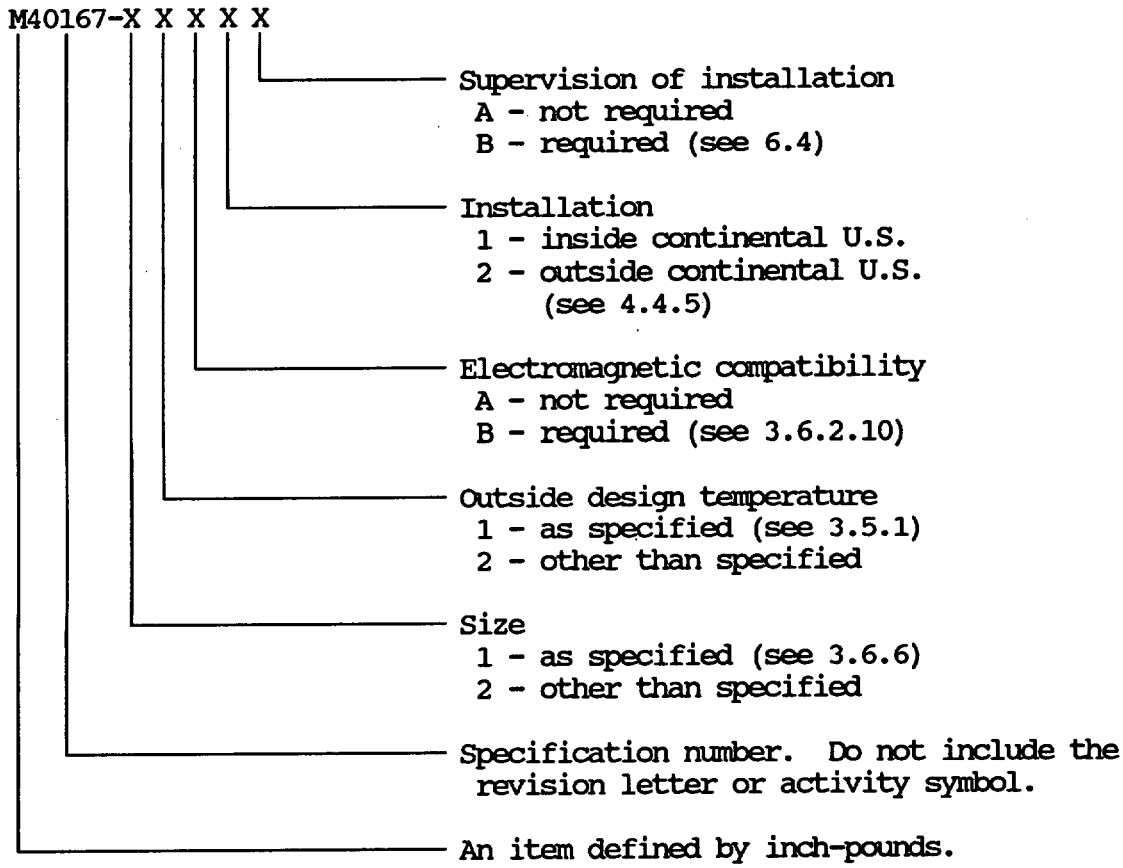
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6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.6 Subject term (key word) listing.

Bread
Container, bakery

6.7 Part or identifying number (PIN) structure. The PINs to be used for items covered by this specification are formed as follows:




6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 84
DIA - GS

User Activity:

Navy - MS

Preparing activity:

Army - GL

Project 7320-0873



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-D-40167D | 92-12-31 |

3. DOCUMENT TITLE
DOUGH PROOFING BOXES, STEAM HEATED (STEAM, OR STEAM AND WATER, HUMIDIFIED)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| U.S. Army Natick RD&E Center | (1) Commercial<br>508-651-4085 | (2) AUTOVON /DSN<br>256-4085 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: STRNC-WEE<br>Natick, MA 01760-5018 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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